

Name:

Student ID:

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Chair of Wireless System
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Winter semester 2025/2026
Cyber Security Application Areas

Exercise sheet 3

Please note:

Submit **your written report** (PDF or Word) and **all used code/scripts** via Moodle by 13 January 2026. The report must include an explanation and proof of concept. Incomplete submissions will not be considered.

The report should clearly explain your entire approach.

Please ensure that your full name and matriculation number are written on all submitted sheets.

The processing of the exercise sheets is obligatory for the participation in the exam.

You must achieve at least 50% of the total points per exercise sheet.

IMPORTANT:

Collaboration in the exercises is considered an attempt to cheat and will be penalized with 0 points for the exercise sheet in question and all parties involved. There will not be a second attempt in this one and you will lose your examination entitlement as a result. The same applies to content plagiarized from the Internet.

Submission date: 13 January 2026

Max. Points: 50

Set the environment

Before starting the setup please download first a VirtualBox from the official website¹.

- Please note that our system is built with the help of the OpenPLC project. So for more technical details and a better understanding of using OpenPLC and ScadaBR, please visit OpenPLC^{2,3}.
- We use the old version of OpenPLC project that is released before Oct 2025.

Download Virtual machine from the Cloud

`https://www.b-tu.de/owncloud/s/aSSw48WZ6WcFBi8`

The password of the machine is 'csaa'.

Import the downloaded .ova file in VirtualBox. If the machine only shows the black screen in the initial phase, try to make the size of the window smaller. The first entering page may not be shown in the full-sized screen.

The OpenPLC runtime has a built-in webserver that you can access. While OpenPLC runtime is running, open your browser and type: **`http://localhost:8080`**

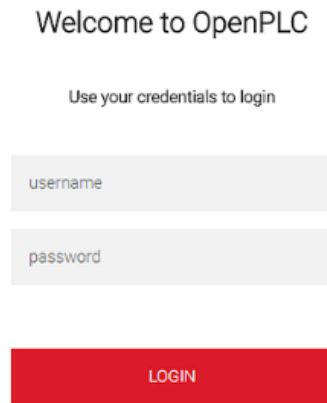
¹ **`https://www.virtualbox.org/wiki/Downloads`**

² **`https://autonomylogic.com/`**

³ **`https://github.com/thiagoralves/OpenPLC\_v3`**

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The image shows a web interface for OpenPLC. At the top, it says "Welcome to OpenPLC". Below that, it says "Use your credentials to login". There are two input fields: one labeled "username" and one labeled "password". Below these fields is a red button labeled "LOGIN".

Welcome to OpenPLC

Use your credentials to login

username

password

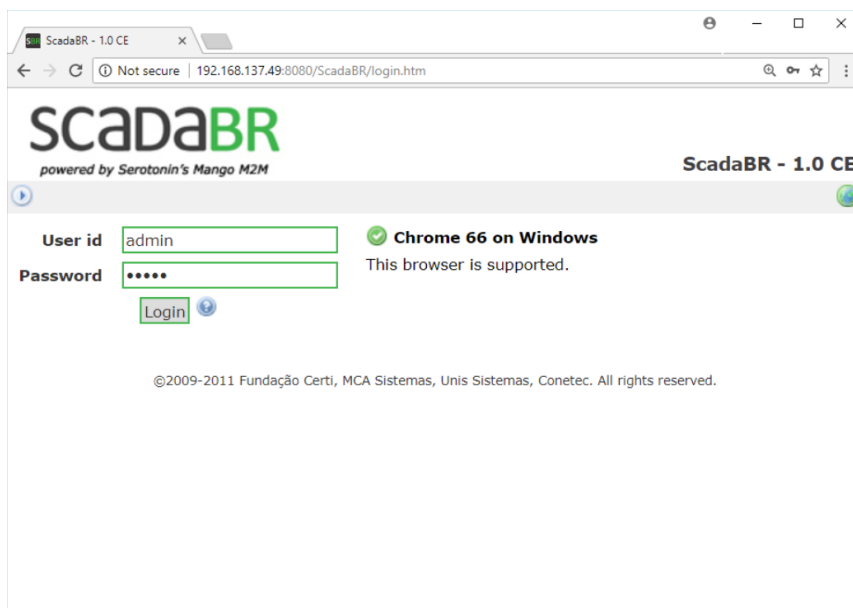
LOGIN

Figure 1: Login page for OpenPLC

Once you access OpenPLC web interface you should see in your browser window a login page like Figure 1. The default username and password is 'openplc' (username) and 'openplc' (password).

If the browser does not load the login page, the OpenPLC service can be manually started by using following commands:

```
sudo systemctl restart openplc
or
cd OpenPLC_v3
./start_openplc.sh
```

The image shows a web browser window displaying the ScadaBR login page. The browser's address bar shows "192.168.137.49:8080/ScadaBR/login.htm". The page has a header with "ScadaBR" in large green letters and "powered by Serotonin's Mango M2M" below it. On the right, it says "ScadaBR - 1.0 CE". The main content area has a "User id" field with "admin" entered, a "Password" field with "*****" entered, and a "Login" button. To the right of the password field, there is a green checkmark icon and the text "Chrome 66 on Windows" and "This browser is supported." At the bottom, there is a copyright notice: "©2009-2011 Fundação Certi, MCA Sistemas, Unis Sistemas, Conetec. All rights reserved.".

ScadaBR - 1.0 CE

powered by Serotonin's Mango M2M

ScadaBR - 1.0 CE

User id: admin

Password: *****

Chrome 66 on Windows
This browser is supported.

Login

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Figure 2: Login page for ScadaBR

Launch your browser and confirm that ScadaBR was installed by going to **http://localhost:9090/ScadaBR**

The default credentials are 'admin' for both username and password.

If the page loads (see Figure 2), then your installation was successful. When you get errors, restart the tomcat service and try again.

```
sudo /opt/tomcat6/apache-tomcat-6.0.53/bin/shutdown.sh
sudo /opt/tomcat6/apache-tomcat-6.0.53/bin/startup.sh
```

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Point Name	Type	Location	Forced	Value
raw_temp	UINT	%IW0	No	0
Ref_Voltage	UINT	%QW0	No	65535
converted_temp	INT	%QW1	No	-29488
voltage	INT	%QW7	No	0
resistance	INT	%QW8	No	0
setpoint	INT	%QW2	No	4000
mode_register	INT	%QW4	No	1
heater_control	INT	%QW5	No	1
heater_man	BOOL	%QX99.0	No	TRUE
heater_auto	BOOL	%QX99.1	No	TRUE
heater	BOOL	%QX0.0	No	TRUE

Figure 3: The control logic in OpenPLC – Monitoring

The given virtual system represents a Water Heater Experiment. It aims at keeping the temperature of a tank at a certain value. Meaning that, if the temperature goes lower than setpoint, the corresponding sensor (Input) reports to the PLC, and the PLC responds by sending a control command to the heater (Output) to switch it ON. The heater remains ON until the temperature is again as high as the configured set point. The control logic program is written in an open source PLC i.e., **OpenPLC** as shown in Figure 3 and works in two functional mode: **Auto** and **Manual**.

The Heater program (water_heater.st) is under /Download/BTUCybersecurity/BTUCyber-security2022/Water Heater Experiment/Ladder Logic/OpenPLC/ folder. Go to the 'Programs' tab in the OpenPLC and upload the water heater program. After uploading the program in OpenPLC, you can see the 'Monitoring' tab as in Figure 3.

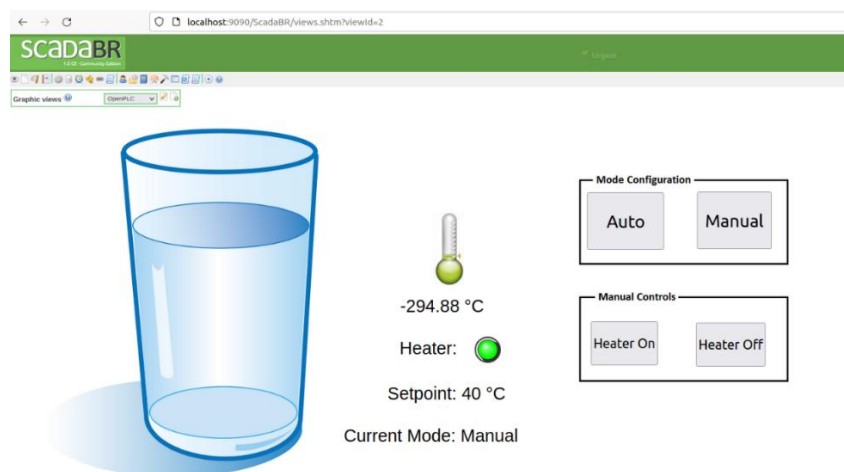


Figure 4: Water Heater Experiment shown in ScadaBR

The user can supervise the entire process using a Human Machine Interface (HMI) built on an open source SCADA system (see Figure 4), called ScadaBR. Both OpenPLC and **ScadaBR** are connected and collaborating together to control and monitor the physical process. With the help of ScadaBR, the user can set a new set point value, change the control mode e.g., from Auto to Manual and the other way around, switch the heater on/off in a manual mode, and so on. To go to the dashboard, click the 'Graphical View' and choose 'OpenPLC'.

* If you have a trouble to retrieve values on ScadaBR, check 'Data Sources' whether the service is receiving the values from 'localhost:502' or reading the holding register correctly. Try to restart the connection with OpenPLC by clicking 'Disable' and 'Enable'.

Tasks:

For the given system, please answer the following questions:

1. In which Security zone do the both parts (PLC and SCADA) run? Which part is considered as a lower trust level? **(Points: 5)**
2. Please perform a real MitM attack against the given system e.g., the attacker places himself in the middle between the OpenPLC and the HMI (ScadaBR), and listens to all the packet transferred between the stations (see Figure 5). In addition, the attacker maliciously alters the packet he receives and sends it back to the other station in order to confuse the physical process. **(Points: 15)**

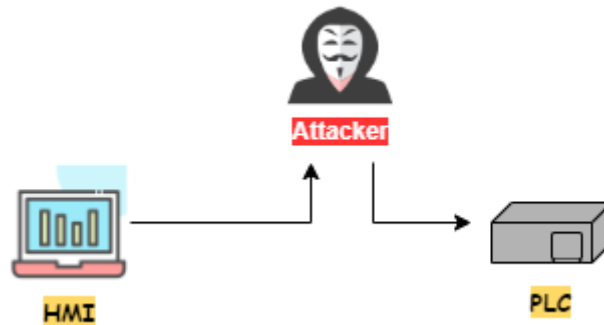


Figure 5: MitM attack scenario

3. Please perform a False Data Injection attack where the attacker sends crafted packets to the target PLC as shown in Figure 6. The attack should change the PLC registers' values, inputs, outputs, etc. At the same time, the user on HMI should not be able to catch your attack. **(Points: 20)**

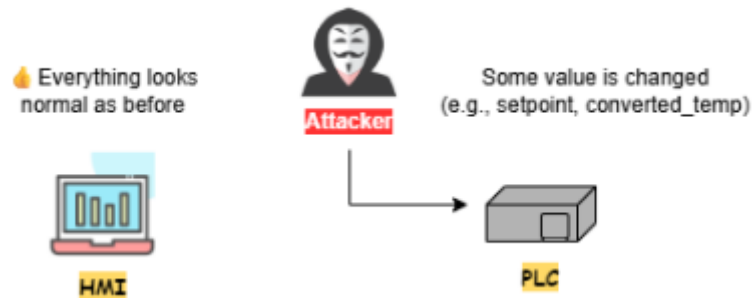


Figure 6: Attacker sends crafted packets to the victim PLC

4. Please perform a flooding attack i.e. Denial of Service (DoS) attack against both HMI and PLC (see Figure 7). The attack should make the PLC/HMI out of the service i.e., the victim is unable to reply to legitimate requests from the user. **(Points: 10)**



Figure 7: Attacker floods the victims with requests/packets